

EDWARD HANISH

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Education

Bachelor of Science: Atmospheric And Earth Science, 05/2026

University of Alabama in Huntsville - Huntsville, AL

- Relevant coursework in Geographic Information Systems, Python, Synoptic and Mesoscale Meteorology, Operational Forecasting.
- Operational Forecasting co-instructed by NWS Huntsville.

Work Experience

Student Researcher, 05/2025 to 12/2025

University of Alabama in Huntsville – Huntsville, AL

- Analyzed Hurricanes Laura and Milton landfall observations using the KLCH and KTBW WSR-88D radar, ASOS data, and FCMP tower measurements to evaluate near-surface wind estimation during high-impact tropical weather.
- Supported analysis of quality-controlled KLCH and KTBW radar data processed through LROSE RadxConvert and Solo3.
- Applied a Py-ART-based Browning Velocity Azimuth Display to analyze horizontal wind profiles from 90–1845 m AGL.
- Incorporated NOAA land-use classification and surface-roughness assumptions to evaluate uncertainty in near-surface wind estimates.

Leadership

UPSTORM Treasurer

August 2025 - May 2026

- Managed club funds and bank accounts.
- Coordinated workshops, led a workshop on radar data focused on the 2022 Gaylord, MI tornado.
- Led and trained students in radiosonde and windsonde weather balloon launches as requested by NWS Huntsville and for outreach events, including Rocket City Weather Fest.

Eagle Scout

February 2019

- Led an installation of a weather station at Hartland Middle School in Hartland, Michigan.
- Collaborated with school staff to determine location, subscription funding, and student use.

Skills

Scientific Python / Data: Python, Py-ART, NumPy, Pandas, matplotlib, Cartopy

Meteorology / Observations: WSR-88D / NEXRAD Level II, Synoptic and Mesoscale Meteorology, Operational Forecasting, Radio/Windsonde Balloon Launches

GIS/Visualization: ArcGIS Pro, Heatmap Analysis, Pairwise Buffer and Proximity Analysis, Point-to-Raster Conversion, Spatial Data QC